

Lecture 1: Introduction

ECON 403/503



Example

V is the value of the company

$u^S = P - V$ where P is the selling price

$$u^B = 1.5V - P$$

Conditional on the transaction being done,
otherwise the utilities are zero

① Everybody knows V

seller will accept $\Leftrightarrow u^S \geq 0$

$$\Leftrightarrow P - V \geq 0$$

$$\Leftrightarrow P \geq V$$

If Buyer is maximizing utility, will offer

$$P_1 = V$$

$$u^S = 0, \quad u^B = 1.5V - V = \frac{V}{2}$$

② V is a R.V. $V \sim U[0, 1]$

$f_V(v) = 1$ pdf of V

Seller will accept $\Leftrightarrow E[u^S] \geq 0$

$$\Leftrightarrow P - E[V] \geq 0$$

$$E[V] = 1/2$$

$$\Leftrightarrow P \geq E[V]$$

If the buyer is max. utility, will offer

$$P_2 = E[V] = 1/2$$

$$u^S = 0, \quad u^B = 1.5 E[V] - P_2$$

$$= 1.5(1/2) - 1/2$$

$$= \frac{3}{2} \left(\frac{1}{2} \right) - \frac{1}{2}$$

$$= \frac{3}{4} - \frac{1}{2} = \frac{1}{4}$$

③ Buyer $V \sim U[0,1]$

Seller knows

Seller accepts $\Leftrightarrow p - v \geq 0$

$$\Leftrightarrow \boxed{p \geq v}$$

The buyer needs to use
this information to update
his beliefs on V

$$V \sim U[0,1] / p \geq v$$

$$u^B = 1.5v - p$$

$$\boxed{E[u^B | p \geq v] = 1.5 E[V | p \geq v] - p}$$

$$E[V | p \geq v]?$$

Fix p

$$Pr(p \geq v) = \int_0^p dv$$

$$= p$$

$$E[V | p \geq v] = \int_0^p \frac{v}{Pr(p \geq v)} dv$$

$$= \int_0^p \frac{v}{p} dv$$

$$= \frac{v^2}{2p} \Big|_0^p = \frac{p}{2}$$

$$E[u^B | p \geq v] = 1.5 \left(\frac{p}{2} \right) - p$$

$$= \frac{3}{2} \left(\frac{p}{2} \right) - p = \frac{3}{4}p - p$$

$$= -\frac{1}{4}p$$

The expected value is, for the buyer

$$-\frac{1}{4}P < 0 \quad \text{for any } P > 0$$

$\Rightarrow$ Buyer will not buy (or offer
a price of 0)
i.e., $P_3 = 0$